Noncommutative homotopy theory

Ulrich Bunke*

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Abstract

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*Fakultät für Mathematik, Universität Regensburg, 93040 Regensburg, ulrich.bunke@mathematik.uni-regensburg.de

1 Overview

1.1 The goals of the course

In this course, we will study C^*-algebras using homotopy theoretic methods. This will allow us to see basic principles of homotopy theory in action in the setting of noncommutative topology. In particular we will see a realization of motivic methods in a specific example.

Remark 1.1. In the following we list parallel examples from various fields with very similar general structure:

field	theories	example	motives
alg. Toplology	Homology theories	$H_{\mathtt{sing}}$	category of spectra $\mathbf{Sp}$
alg. geometry	Weil cohomology theories	H_{et}	stable mot. homot. cat. HS
coarse geometry	coarse homology theories	$K\mathcal{X}$	coarse motivic $\mathbf{Sp}\mathcal{X}$
stable ∞-categories	localizing invariants	K^{alg}	univ. localizing invariant $\mathcal{U}_{\mathrm{loc}}$
C^*-algebras	homological functors	K^{top}	KK and E-theory

In the following, we indicate the meaning of the word *noncommutative*. Geometric objects are often studied through their algebras of functions: regular functions in algebraic geometry, smooth functions in differential geometry, and continuous functions in topology. These algebras are commutative. Noncommutative geometry extends constructions and invariants from commutative to general associative algebras which are interpreted as representatives of *noncommutative spaces*.

A further principle is Morita invariance: a construction or an invariant should depend on an appropriate category of modules over an algebra rather than on that algebra itself. This principle leads to modelling spaces through appropriate categories of modules and their generalizations, and to phrase constructions and invariants using categorical language.

We will study an instance of this idea in topology. Gelfand–Naimark duality identifies compact Hausdorff spaces with commutative unital C^*-algebras, contravariantly, via their continuous functions. The category of commutative C^*-algebras embeds into the category of all C^*-algebras whose objects are henceforth interpreted as noncommutative spaces. Morita invariance comes in through $\mathbb{K}$-stability: constructions and invariants should be invariant under tensoring with the C^*-algebra of compact operators.

Example 1.2. Let X be a finite set and G be a finite group acting on X. We compare the limit of the G-action in topology with a noncommutative version in which, after applying Gelfand duality, equality is weakened to unitary conjugacy.

The data gives the algebra $C(X) := \bigoplus_X \mathbb{C}$ generated by the components δ_x for x in X with the action of G on the index set. In topology, we would consider the fixed point

set $X^G := \mathtt{lim}_{BG}\, X$ and the inclusion $X^G \to X$ inducing the projection $C(X) \to C(X^G)$. Since Gelfand duality is contravariant, it switches limits and colimits and we would get $C(X^G) \cong \mathtt{colim}_{BG}\, C(X) =: C(X)/G$. This makes δ_x equal to δ_{gx}. If for example the action of G on X is free, then

$$C(X)/G \cong C(X^G) \cong C(\emptyset) \cong 0 \ .$$

If $X = *$, then

$$C(X)/G \cong C(X^G) \cong C(X) \ .$$

The proposal of non-commutative topology is not to make δ_x equal to δ_{gx}, but to add a unitary ρ_g for every g in G such that $\rho_g \delta_x \rho_g^{-1} = \delta_{gx}$ and $\rho_g \rho_h = \rho_{gh}$. The algebra $C(X) \rtimes G$ generated by $C(X)$ and the ρ_g for all g in G subject to these relations is called the crossed product. It retains more information than $C(X^G)$.

For example, if the action is free, then

$$C(X) \rtimes G \cong \bigoplus_{X/G} \mathtt{Mat}_{|G|}(\mathbb{C}) \ .$$

If $X = *$, then

$$C(*) \rtimes G \cong C^*(G) \cong \bigoplus_{\rho \in \hat{G}} \mathtt{Mat}_{\dim(\rho)}(\mathbb{C}) \ ,$$

where $\hat{G}$ is the dual of G, the set of isomorphism classes of finite-dimensional representations. The following schematically pictures what we have done above:

commutative colimit	add unitaries ρ_g
force $g(a) = a$	$g(a) = \rho_g a \rho_g^*$
$\Downarrow$	$\Downarrow$
$C(X^G)$	$C(X) \rtimes G$
colimit in $*\mathbf{Alg}$	colimit in $C^*\mathbf{Cat}[W^{-1}]$

The last line indicates the categorical setup where one can interpret adding the unitaries as taking a categorical quotient. In the case above, this happens in the 2-category $C^*\mathbf{Cat}[W^{-1}]$ which is obtained from the 1-category $C^*\mathbf{Cat}$ of C^*-categories by inverting unitary equivalences. □

This leads us to explain the general principles behind the notion of *homotopy theory*. Typically one starts with a category $\mathcal{C}$ and chooses a class of morphisms W, called *weak equivalences*. One then universally inverts them by a Dwyer–Kan localization,

$$\ell : \mathcal{C} \longrightarrow \mathcal{C}[W^{-1}]$$

whose precise definition requires the language of ∞-categories. Unlike an ordinary localization, it retains mapping spaces and their higher homotopies.

Example 1.3. For example, localizing the category **CW** of CW-complexes at homotopy equivalences gives the ∞-category **Ani** of *anima*, or homotopy types. □

Doing homotopy theory is then understood as studying constructions with objects from $\mathcal{C}$ and their interpretation in $\mathcal{C}[W^{-1}]$. Typical examples are homotopy limits or colimits.

Example 1.4. Let $A : I \to \mathcal{C}$ be a diagram. A cone $\tilde{A} : I^{\triangleright} \to \mathcal{C}$ with tip $A(\infty)$ is called *homotopy colimit cone*, if the $\ell(\tilde{A})$ is a colimit cone in $\mathcal{C}[W^{-1}]$, i.e.,

$$\ell A(\infty) \simeq \operatorname*{colim}_{I} \ell A \ .$$

So a homotopy colimit is data in $\mathcal{C}$ which presents a colimit in the localization.

The same reasoning applies to homotopy limits. □

Example 1.5. Let $\mathbf{Cat}_{\mathbb{C}}$ be the 1-category of $\mathbb{C}$-linear categories and linear functors. The localization

$$\ell : \mathbf{Cat}_{\mathbb{C}} \to \mathbf{Cat}_{\mathbb{C}}[W^{-1}]$$

at the equivalences produces the 2-category of $\mathbb{C}$-linear categories, functors and isomorphisms.

A unital algebra A can be viewed as a $\mathbb{C}$-linear category with a single object. If a group G acts on A, then the (algebraic) crossed product fits into

$$\ell(A \rtimes G) \simeq \operatorname*{colim}_{BG} \ell(A) \ .$$

So $A \rtimes G$ is the homotopy colimit of A with respect to the G-action. □

In the course, we will introduce the language of ∞-categories as it is needed. We emphasize the applications and take the internal theory as granted.

In this course we will start from the category $C^*\mathbf{Alg}^{\mathrm{nu}}$ of C^*-algebras. Different classes of weak equivalences in $C^*\mathbf{Alg}^{\mathrm{nu}}$ lead to different homotopy theories. Our principal examples will be $\mathbb{K}$-stabilized homotopy theory, which we call Morita theory, and the categories KK and E representing KK- and E-theory. Understanding their categorical properties will require both homotopy theoretic methods and results about C^*-algebras. We will develop these ingredients step by step.

We start the course with an introduction of the category $C^*\mathbf{Alg}^{\mathrm{nu}}$ of complex, possibly non-unital C^*-algebras and possibly non-unital homomorphisms in a way which makes its categorical properties visible as quickly as possible.

Under Gelfand duality, its full subcategory of commutative algebras is equivalent to the category $(\mathrm{LCH}^+)^{\mathrm{op}}$, where LCH^+ denotes locally compact Hausdorff spaces and proper continuous maps defined on open subsets. The inclusion

$$\mathrm{LCH}^+ \hookrightarrow C^*\mathbf{Alg}^{\mathrm{nu}\,\mathrm{op}}$$

is the extension of the category of classical spaces by non-commutative spaces.

The main motivating invariant of non-commutative spaces is topological K-theory,

$$K : C^*\mathbf{Alg}^{\mathrm{nu}} \longrightarrow \mathbf{Sp},$$

where $\mathbf{Sp}$ is the ∞-category of spectra. It extends the classical topological K-theory of locally compact Hausdorff spaces. Its basic properties are homotopy invariance, $\mathbb{K}$-stability, exactness, and compatibility with filtered colimits. They motivate the introduction of the localizations

$$\mathrm{kk} : C^*\mathbf{Alg}^{\mathrm{nu}} \to \mathrm{KK}\ , \qquad \mathrm{e} : C^*\mathbf{Alg}^{\mathrm{nu}} \to \mathrm{E}$$

implementing these properties universally. The difference between kk and e lies the details of the exactness condition.

The categories KK and E capture the motives of noncommutative spaces.

Essentially by definition, we obtain

$$K(A) \simeq \mathrm{map}_{\mathrm{KK}}(\mathrm{kk}(\mathbb{C}), \mathrm{kk}(A)) \simeq \mathrm{map}_{\mathrm{E}}(\mathrm{e}(\mathbb{C}), \mathrm{e}(A)).$$

Here map denotes the mapping *spectrum*. We will use this definition of K-theory to derive its properties and to compute the groups $K_i(A) := \pi_i K(A)$ in examples.

The following subsections outline the steps towards these constructions which will be the guiding principle for this course. Theorems stated without proof here will, of course, be established later in the course.

1.2 The category of C^*-algebras

For this overview we use a concrete characterization of C^*-algebras, which we will prove equivalent to the definition used in Section 2. We start with the algebraic data.

Definition 1.6. *A complex* $*$-algebra *is an associative, possibly non-unital, complex algebra A with an antilinear involution $a \mapsto a^*$ satisfying $(ab)^* = b^*a^*$. A morphism is a complex-linear, multiplicative map preserving the involution; it need not preserve a unit. We denote their category by* $*\mathbf{Alg}^{\mathrm{nu}}$.

Example 1.7. The algebra $B(H)$ of bounded operators on a complex Hilbert space H is a complex $*$-algebra, with composition as multiplication and the adjoint as involution.

Another example is $C(\bar{B}^n)$, the algebra of continuous complex-valued functions on the closed unit ball $\bar{B}^n \subseteq \mathbb{R}^n$, with pointwise operations and $f^* = \bar{f}$.

Every $*$-invariant subalgebra of a $*$-algebra is again a complex $*$-algebra. Examples include the finite-rank operators in $\mathbb{B}(H)$ and the restrictions of complex polynomials in the coordinate functions to $\bar{B}^n$, or $C_c(B^n)$, regarded as a subalgebra of $C(\bar{B}^n)$ by extension by zero from the open ball B^n. These subalgebras need not be norm-closed.

On $\mathtt{Mat}_2(\mathbb{C})$, let $J = \mathrm{diag}(1, -1)$ and define $a^\dagger := Ja^*J$, where a^* is the conjugate transpose. Since $J^* = J$ and $J^2 = 1$, this operation is antilinear and satisfies

$$(a^\dagger)^\dagger = a, \qquad (ab)^\dagger = b^\dagger a^\dagger.$$

It therefore defines another complex $*$-algebra structure on $\mathtt{Mat}_2(\mathbb{C})$.

□

Definition 1.8. *A* C^*-algebra *is a complex $*$-algebra that is isomorphic, as a $*$-algebra, to a norm-closed $*$-invariant subalgebra of $B(H)$ for some complex Hilbert space H. The category $C^*\mathbf{Alg}^{\mathrm{nu}}$ is the full subcategory of $*\mathbf{Alg}^{\mathrm{nu}}$ consisting of C^*-algebras.*

Example 1.9. The algebra $K(H)$ of compact operators is the norm closure in $B(H)$ of the finite-rank operators, so it is a C^*-algebra. If H is infinite-dimensional, $K(H)$ is non-unital. We write $\mathbb{K} := K(\ell^2)$.

If X is a compact Hausdorff space, then $C(X)$ is a C^*-algebra. To see this directly from the definition, represent it on $\ell^2(X)$ by diagonal multiplication:

$$(\rho(f)\xi)(x) = f(x)\xi(x).$$

Since $\|\rho(f)\| = \sup_{x\in X} |f(x)|$, this representation is isometric. Completeness of $C(X)$ in the supremum norm shows that its image is closed.

In contrast, $\mathtt{Mat}_2(\mathbb{C})$ with the involution $a^\dagger = Ja^*J$ is not a C^*-algebra (see exercise). □

We will observe the following categorical properties:

Theorem 1.10. *The category $C^*\mathbf{Alg}^{\mathrm{nu}}$ is pointed, with zero object the zero algebra, and admits all small limits and colimits. It is $\aleph_1$-presentable and the subcategory of $\aleph_1$-compact objects $(C^*\mathbf{Alg}^{\mathrm{nu}})^{\aleph_1}$ is the subcategory of separable C^*-algebras.*

Example 1.11. The product of two C^*-algebras A and B is $A \oplus B$, with coordinatewise operations. The coproduct is their free product $A * B$ in the non-unital category; no relation identifies their units when the factors happen to be unital. For instance, $\mathbb{C} * \mathbb{C}$ is generated by two projections with no commutation relation imposed between them. □

The category $C^*\mathbf{Alg}^{\mathrm{nu}}$ also carries two canonical symmetric monoidal structures, with tensor products

$$A \otimes_{\max} B \qquad \text{and} \qquad A \otimes_{\min} B$$

and tensor unit $\mathbb{C}$. They are characterized by universal properties:

1. If $A \subseteq \mathbb{B}(H)$ and $B \subseteq \mathbb{B}(L)$, then $A \otimes_{\min} B \subseteq \mathbb{B}(H \otimes L)$ is the norm closure of the represented algebraic tensor product.

2. The maximal tensor product has the following universal property: for every C^*-algebra C and pair of $*$-homomorphisms $f : A \to C$ and $g : B \to C$ with commuting images, the $*$-homomorphism $A \otimes^{\mathrm{alg}} B \to C$ given by $a \otimes b \mapsto f(a)g(b)$ extends uniquely to a $*$-homomorphism $A \otimes_{\max} B \to C$.

These are completions of the algebraic tensor product in generally different C^*-norms.

1.3 Gelfand duality

Gelfand duality identifies the opposite of LCH^+ with the full subcategory $C^*\mathbf{Alg}^{\mathrm{nu}}_{\mathrm{comm}}$ of commutative C^*-algebras. Its simplest formulation concerns compact spaces. Let $\mathbf{CH}$ be the category of compact Hausdorff spaces and continuous maps, and let $\mathbf{C}^*\mathbf{Alg}_{\mathrm{comm}}$ be the category of commutative *unital* C^*-algebras and unital homomorphisms. Here the zero algebra is allowed as a unital algebra, corresponding to the empty space.

Theorem 1.12. *Taking the algebra of continuous $\mathbb{C}$-valued functions induces an equivalence*

$$C(-) : \mathbf{CH}^{\mathrm{op}} \xrightarrow{\simeq} \mathbf{C}^*\mathbf{Alg}_{\mathrm{comm}}.$$

The inverse associates to A the space $\mathtt{Hom}_{\mathbf{C}^*\mathbf{Alg}}(A, \mathbb{C})$ of characters with the topology of pointwise convergence.

For $C(X)$ the characters are exactly evaluations at points of X.

For locally compact Hausdorff X, one uses $C_0(X)$, the continuous functions vanishing at infinity. A proper map $f : U \to Y$, with $U \subseteq X$ open, induces $C_0(Y) \to C_0(X)$ by pullback to U followed by extension by zero. Every homomorphism between commutative C^*-algebras arises uniquely in this way.

Every complex algebra A admits a unitalization A^u with underlying vector space $A \oplus \mathbb{C}$, unit $(0,1)$, and multiplication

$$(a,\lambda)(b,\mu) = (ab + \lambda b + \mu a, \lambda\mu).$$

Definition 1.13. *For an element a of a unital algebra A, its* spectrum *is*

$$\sigma_A(a) := \{\lambda \in \mathbb{C} \mid a - \lambda 1_A \text{ is not invertible in } A\}.$$

If A is non-unital, we use the same definition in its unitization A^u, identifying a with $(a,0)$. We write $\sigma(a)$ when the ambient algebra is clear.

For every element of a nonzero complex Banach algebra, the spectrum is nonempty and compact. If A is a C^*-algebra and $a = a^*$, then $\sigma(a) \subseteq \mathbb{R}$.

Gelfand duality gives the *continuous functional calculus*: for selfadjoint a in a unital C^*-algebra A, there is a unique unital $*$-homomorphism

$$\Phi_a : C(\sigma(a)) \longrightarrow A, \qquad f \longmapsto f(a),$$

sending the coordinate function to a. Its image is the unital C^*-subalgebra generated by a. In the non-unital case this homomorphism initially takes values in A^u; the subalgebra

$$\{f \in C(\sigma(a)) \mid f(0) = 0\} \cong C_0(\sigma(a) \setminus \{0\})$$

maps into A.

Example 1.14. Suppose first that $A \subseteq \mathbb{B}(H)$ is unital with the same identity, and a in A is selfadjoint. The spectral theorem associates a projection $E_a(S) \in \mathbb{B}(H)$ to every Borel subset S of $\mathbb{R}$. Such a projection need not belong to A.

If a subset S of $\sigma(a)$ is both open and closed, however, its characteristic function is continuous, and

$$E_a(S) = \chi_S(a) = \Phi_a(\chi_S) \in A.$$

This applies, for example, to a component separated from the rest of the spectrum by a gap. For non-unital A, the same conclusion holds for a clopen spectral subset not containing zero, by the vanishing condition above. □

Example 1.15. Here is a typical application of functional calculus to evolution equations

$$\partial_t u + \Delta u = 0, \qquad u(0) = u_0,$$

where Δ is a possibly unbounded nonnegative selfadjoint operator on a Hilbert space H. For $u_0 \in H$, spectral calculus defines

$$u(t) = e^{-t\Delta} u_0, \qquad t \geq 0.$$

Defining $e^{-t\Delta}$ requires the spectral calculus for unbounded selfadjoint operators. This extension will be discussed in the parallel course on *Dirac operators in coarse geometry*. □

1.4 Inverting homotopy equivalences

The category $C^*\mathbf{Alg}^{\mathrm{nu}}$ is enriched in topological spaces. We equip its spaces of homomorphisms with the compactly generated version of the point-norm topology such that for every compact Hausdorff space W there is a natural bijection

$$\mathtt{Hom}_{\mathbf{Top}}(W, \mathtt{Hom}_{C^*\mathbf{Alg}^{\mathrm{nu}}}(A, B)) \cong \mathtt{Hom}_{C^*\mathbf{Alg}^{\mathrm{nu}}}(A, C(W) \otimes B).$$

A homotopy between $f_0, f_1 : A \to B$ is a path in this mapping space. A homomorphism is a homotopy equivalence if it has a two-sided inverse up to homotopy.

We let W_h denote the class of homotopy equivalences. Our first localization is

$$L_h : C^*\mathbf{Alg}^{\mathrm{nu}} \longrightarrow C^*\mathbf{Alg}^{\mathrm{nu}}_h := C^*\mathbf{Alg}^{\mathrm{nu}}[W_h^{-1}].$$

Its universal property says that functors out of $C^*\mathbf{Alg}^{\mathrm{nu}}_h$ to arbitrary ∞-categories are the same as functors out of $C^*\mathbf{Alg}^{\mathrm{nu}}$ which send homotopy equivalences to equivalences. Let

$$\ell : \mathbf{Top} \to \mathbf{Ani}$$

be the canonical functor sending a topological space to its weak homotopy type.

Theorem 1.16. *The ∞-category $C^*\mathbf{Alg}^{\mathrm{nu}}_h$ is pointed and left-exact and admits all coproduct. For A, B in $C^*\mathbf{Alg}^{\mathrm{nu}}$ we have*

$$\mathtt{Hom}_{C^*\mathbf{Alg}^{\mathrm{nu}}_h}(L_h(A), L_h(B)) \simeq \ell\mathtt{Hom}_{C^*\mathbf{Alg}^{\mathrm{nu}}}(A, B) \ .$$

We will determine which limits and colimits L_h preserves, using a category of fibrant objects structure on $C^*\mathbf{Alg}^{\mathrm{nu}}$. In particular, a categorical pullback of algebras does not automatically become a homotopy pullback after localization.

Example 1.17. The following are equivalences in $C^*\mathbf{Alg}^{\mathrm{nu}}_h$, i.e., after applying L_h:

1. the inclusion $\mathbb{C} \to C(\bar{B}^n)$
2. the inclusion $0 \to C_0([0, \infty))$
3. the map $1 \otimes \epsilon : \mathbb{K} \cong \mathbb{K} \otimes \mathbb{C} \to \mathbb{K} \otimes \mathbb{K}$ for $\epsilon : \mathbb{C} \to \mathbb{K}$ such that $1 \mapsto p$ with a one-dimensional projection p

Furthermore, the upper left and lower right corner inclusions $\mathbb{C} \to \mathtt{Mat}_2(\mathbb{C})$ become equivalent after applying L_h. □

1.5 Morita invariance

For C^*-algebras, Morita invariance is introduced using $\mathbb{K}$-stabilization. For concreteness $\mathbb{K} := \mathbb{K}(\ell^2)$.

Definition 1.18. *A morphism $f : A \to B$ is a $\mathbb{K}$-equivalence if*

$$f \otimes \mathtt{id}_{\mathbb{K}} : A \otimes \mathbb{K} \longrightarrow B \otimes \mathbb{K}$$

is a homotopy equivalence.

Example 1.19. $\mathbb{C} \to \mathbb{K}$ with $1 \mapsto p$ with $\dim(p) = 1$ is a $\mathbb{K}$-equivalence. □

Remark 1.20. This should not be confused with the notion of *stable isomorphism*: Two C^*-algebras are stably isomorphic if there exists an isomorphism $A \otimes \mathbb{K} \cong B \otimes \mathbb{K}$.

Let $W_{h,K}$ be the class of $\mathbb{K}$-equivalences. It already contains W_h, since tensoring preserves homotopies. The localization

$$L_{K,h} : C^*\mathbf{Alg}^{\mathrm{nu}} \longrightarrow L_K C^*\mathbf{Alg}^{\mathrm{nu}}_h := C^*\mathbf{Alg}^{\mathrm{nu}}[W_{h,K}^{-1}]$$

will be called Morita theory or $\mathbb{K}$-stabilized homotopy theory. A functor inverting homotopy equivalences and $\mathbb{K}$-equivalences is called homotopy invariant and $\mathbb{K}$-stable. Since $\mathbb{K} \otimes \mathtt{Mat}_n(\mathbb{C}) \cong \mathbb{K}$, such a functor is also matrix stable. By definition, $L_{K,h}$ is the universal homotopy invariant and $\mathbb{K}$-stable functor.

Theorem 1.21. *We have a left Bousfield localiztion*

$$L_K := \mathbb{K} \otimes - : C^*\mathbf{Alg}^{\mathrm{nu}}_h \leftrightarrows L_K C^*\mathbf{Alg}^{\mathrm{nu}}_h : \mathrm{incl} \ .$$

The ∞-category $L_K C^*\mathbf{Alg}^{\mathrm{nu}}_h$ is left-exact and semi-additive, and admits all small coproducts.

It follows that

$$\mathtt{Hom}_{L_K C^*\mathbf{Alg}^{\mathrm{nu}}_h}(L_{K,h}(A), L_{K,h}(B)) \simeq \ell\mathtt{Hom}_{C^*\mathbf{Alg}^{\mathrm{nu}}}(A, B \otimes \mathbb{K}) \ .$$

Semi-additivity means that finite products and coproducts agree via the canonical comparison map. Every object in a semiadditive category naturally refines to a commutative monoid.

$\mathbb{K}$-stabilization is a specifically noncommutative operation.

Example 1.22. Let $\eta_n : S^{n+1} \to S^n$, $n \geq 3$, be a suspension of the Hopf map, and put $X = C\eta_n$ and $Y = S^n \vee S^{n+2}$. Write $C_*(Z) := \{f \in C(Z) \mid f(*) = 0\}$ for the functions vanishing at the base point. Dadarlat–McClure's show that $C_*(X) \otimes \mathbb{K}$ and $C_*(Y) \otimes \mathbb{K}$ are homotopy equivalent.[1] On the other hand, Sq^2 distinguishes X and Y up to homotopy equivalence. Consequently $C_*(X)$ and $C_*(Y)$ are equivalent in $L_K C^*\mathbf{Alg}_h^{\mathrm{nu}}$, but not in $C^*\mathbf{Alg}_h^{\mathrm{nu}}$.

□

1.6 Excision, exactness, and group objects

In topology, excision gives rise to Mayer–Vietoris sequences. Extensions of C^*-algebras provide an algebraic counterpart.

Definition 1.23. *An* exact sequence *in* $C^*\mathbf{Alg}^{\mathrm{nu}}$ *is a commutative square*

$$\begin{array}{ccc} I & \xrightarrow{j} & A \\ \downarrow & & \downarrow{\scriptstyle q} \\ 0 & \longrightarrow & Q \end{array}$$

which is bicartesian.

Definition 1.24. *Let $\mathcal{D}$ be a pointed ∞-category with finite limits. A functor $F : C^*\mathbf{Alg}^{\mathrm{nu}} \to \mathcal{D}$ is* excisive *if it is reduced, meaning $F(0) \simeq 0$, and sends exact sequences to pullback squares.*

For $\mathcal{D} = \mathbf{Sp}$, this yields

$$\cdots \longrightarrow \pi_i F(I) \longrightarrow \pi_i F(A) \longrightarrow \pi_i F(Q) \xrightarrow{\partial} \pi_{i-1} F(I) \longrightarrow \cdots$$

for all integers i.

Example 1.25. Let $F : C^*\mathbf{Alg}^{\mathrm{nu}} \to \mathbf{Sp}$ be excisive. Suppose that a compact Hausdorff space X is the union of closed subsets A and B, and put $D := A \cap B$. Restriction of functions gives a pullback square

$$\begin{array}{ccc} C(X) & \xrightarrow{r_A} & C(A) \\ {\scriptstyle r_B}\downarrow & & \downarrow{\scriptstyle s_A} \\ C(B) & \xrightarrow[s_B]{} & C(D). \end{array} \tag{1.1}$$

[1] M. Dadarlat and J. McClure, *When are two commutative C^*-algebras stably homotopy equivalent?*, Math. Z. 235 (2000), 499–523, Theorem 3.5 and Proposition 6.2, https://doi.org/10.1007/s002090000143.

Indeed, continuous functions on A and B agreeing on D glue uniquely to a continuous function on X. The restriction maps are surjective by the Tietze extension theorem.

Set $U := X \setminus B = A \setminus D$. The kernels of r_B and s_A are both canonically isomorphic to the ideal $C_0(U)$, using extension by zero into X and into A, respectively. We therefore have the following commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & C_0(U) & \xrightarrow{j_X} & C(X) & \xrightarrow{r_B} & C(B) \longrightarrow 0 \\ & & \Vert{\scriptstyle \mathtt{id}} & & \downarrow{\scriptstyle r_A} & & \downarrow{\scriptstyle s_B} \\ 0 & \longrightarrow & C_0(U) & \xrightarrow{j_A} & C(A) & \xrightarrow{s_A} & C(D) \longrightarrow 0. \end{array} \tag{1.2}$$

Applying F produces a map of fibre sequences and

$$\begin{array}{ccc} F(C(X)) & \xrightarrow{F(r_A)} & F(C(A)) \\ {\scriptstyle F(r_B)}\downarrow & & \downarrow{\scriptstyle F(s_A)} \\ F(C(B)) & \xrightarrow[F(s_B)]{} & F(C(D)) \end{array}$$

is a pullback square.

We get a commutative diagram of long exact sequences

$$\begin{array}{ccccccccccc} \cdots \to \pi_i F(C_0(U)) & \to & \pi_i F(C(X)) & \to & \pi_i F(C(B)) & \xrightarrow{\partial_{X,*}} & \pi_{i-1} F(C_0(U)) & \longrightarrow \cdots \\ \Vert{\scriptstyle \mathtt{id}} & & \downarrow{\scriptstyle r_{A,*}} & & \downarrow{\scriptstyle s_{B,*}} & & \Vert{\scriptstyle \mathtt{id}} & \\ \cdots \to \pi_i F(C_0(U)) & \to & \pi_i F(C(A)) & \to & \pi_i F(C(D)) & \xrightarrow{\partial_{A,*}} & \pi_{i-1} F(C_0(U)) & \to \cdots . \end{array}$$

Taking homotopy groups gives the Mayer–Vietoris sequence

$$\begin{aligned} \cdots \longrightarrow \pi_i F(C(X)) \xrightarrow{(r_{A,*}, r_{B,*})} \pi_i F(C(A)) \oplus \pi_i F(C(B)) \xrightarrow{s_{A,*} - s_{B,*}} \pi_i F(C(D)) \\ \xrightarrow{\delta_*} \pi_{i-1} F(C(X)) \longrightarrow \cdots . \end{aligned} \tag{1.3}$$

Here $\delta_* = F(j_X)_* \circ \partial_{A,*}$. Taking $F = K$ recovers the Mayer–Vietoris sequence in operator algebra K-theory, or, contravariantly on spaces, in topological K-cohomology. □

The example illustrates why extensions are the right input for excision. We now impose this condition universally. There is a left-exact localization of the $\mathbb{K}$-stabilized homotopy category

$$L_{\mathrm{ex}} : L_K C^* \mathbf{Alg}_h^{\mathrm{nu}} \longrightarrow \mathcal{C}_{\mathrm{ex}}.$$

Its composite $L := L_{\mathrm{ex}} \circ L_{K,h}$ is universal among homotopy invariant, $\mathbb{K}$-stable, excisive and Schochet-exact functors to pointed ∞-categories with finite limits, with finite-limit-preserving functors between the targets. Here Schochet exact means that that functor sends Schochet fibrant squares to cartesian squares.

If $W_{\mathrm{ex},K,h}$ denotes the class of maps of algebras inverted by L, we can also write

$$\mathcal{C}_{\mathrm{ex}} \simeq C^*\mathbf{Alg}^{\mathrm{nu}}[W_{\mathrm{ex},K,h}^{-1}].$$

The category $\mathcal{C}_{\mathrm{ex}}$ is still left-exact and semi-additive. Every object therefore has a canonical commutative monoid structure. An object M is a *group object* if this structure is grouplike. Equivalently, for every T, the commutative monoid $\pi_0 \operatorname{Map}(T, M)$ is a group. We write $\mathcal{C}_{\mathrm{ex}}^{\mathrm{group}}$ for the full subcategory of these objects.

Passing to group objects has a particularly concrete description here. The Toeplitz extension produces a natural Bott map

$$\beta_M : \Omega^2 M \longrightarrow M.$$

Bott periodicity asserts that this map is an equivalence for group objects. In addition, $\Omega^2 M$ is always a group object. These facts lead to the following result.

Theorem 1.26. *There is an adjunction*

$$\mathrm{incl} : \mathcal{C}_{\mathrm{ex}}^{\mathrm{group}} \rightleftarrows \mathcal{C}_{\mathrm{ex}} : \Omega^2 \ ,$$

whose left adjoint is the fully faithful inclusion and whose counit is $\beta : \mathrm{incl} \circ \Omega^2 \to \mathtt{id}$. *Thus the group objects form a right Bousfield localization of* $\mathcal{C}_{\mathrm{ex}}$. *The category* $\mathcal{C}_{\mathrm{ex}}^{\mathrm{group}}$ *is stable.*

These constructions are the essential intermediate steps in the construction of the E-theory functor

$$\mathrm{e} : C^*\mathbf{Alg}^{\mathrm{nu}} \to \mathrm{E}$$

as the universal functor to a cocomplete stable ∞-category that is homotopy invariant, $\mathbb{K}$-stable, excisive, countable sum-preserving, $\aleph_1$-filtered colimit preserving. For

$$\mathrm{kk} : C^*\mathbf{Alg}^{\mathrm{nu}} \to \mathrm{KK}$$

we require a slightly weaker excisiveness condition: it is only required that semisplit sequences are sent to fibre sequences.

1.7 Outlook

The constructions above concern C^*-algebras without additional symmetry. Many geometric and analytic applications instead involve algebras with an action of a group G.

Equivariant homotopy invariance, stabilization and exactness lead to equivariant versions of KK- and E-theory. A new feature is functoriality in the group: restriction, induction and their adjunctions relate theories for different groups. Crossed products provide a further link, turning an algebra with a group action into an ordinary C^*-algebra. Understanding how these operations pass through the localizations is a central part of the equivariant theory.

These constructions connect harmonic analysis with topology and index theory. In particular, for a discrete group G, the Baum–Connes assembly map relates equivariant topological K-homology to the K-theory of the reduced group C^*-algebra $C_r^*(G)$. Equivariant KK-theory provides a framework for constructing this map and studying the consequencesof the Baum-Connes conjecture. We plan to develop the equivariant theory in the continuation of this course in the summer semester of 2027.

2 The category of C^*-algebras

We introduce the categories of complex $*$-algebras, pre-C^*-algebras and C^*-algebras

$$C^*\mathbf{Alg}^{\mathrm{nu}} \subseteq C^*_{\mathrm{pre}}\mathbf{Alg} \subseteq *\mathbf{Alg}^{\mathrm{nu}}.$$

All algebras are complex and may be non-unital, and all homomorphisms may fail to preserve units.

2.1 $*$-algebras

Definition 2.1. *A complex* $*$-algebra *is a complex vector space A with an associative bilinear multiplication and an antilinear map $a \mapsto a^*$ satisfying*

$$(a^*)^* = a, \qquad (ab)^* = b^*a^*.$$

A $*$-homomorphism *is a complex-linear, multiplicative map commuting with involution. These objects and morphisms form the category $*\mathbf{Alg}^{\mathrm{nu}}$.*

So, the existence of units is not required, and if an algebra happens to have a unit, a morphism is still not required to preserve it.

Example 2.2. The zero algebra is both initial and terminal in $*\mathbf{Alg}^{\mathrm{nu}}$, so this category is pointed. □

Example 2.3. An ideal will mean a complex-linear two-sided $*$-invariant ideal. It is a complex $*$-algebra by itself. Furthermore, the quotient A/J inherits a complex $*$-algebra structure. □

Example 2.4. We have the following examples of complex $*$-algebras and ideals:

1. The algebra $\mathbb{B}(H)$ of bounded operators on a Hilbert space with A^* being the adjoint of A. The compact operators $\mathbb{K}(H)$ and the finite rank operators $\mathbb{B}(H)^{\mathrm{fin-rk}}$ are $*$-ideals.

2. The algebra $C_0(X)$ for a locally compact Hausdorff space with $f^* := \bar{f}$. For an open subset U of X we have the ideal $C_0(U) \subseteq C_0(X)$, embedded by extension by zero.

3. The algebra $C^\infty(M)$ of smooth functions on a manifold with $f^* := \bar{f}$.

4. The polynomial algebra $\mathbb{C}[t]$ with involution determined by $t^* = t$. Thus $(\sum_k \lambda_k t^k)^* = \sum_k \bar{\lambda}_k t^k$.

5. For a discrete group G, the group algebra $\mathbb{C}[G]$ consists of finite sums $\sum_g \lambda_g g$, with involution $(\sum_g \lambda_g g)^* = \sum_g \bar{\lambda}_g g^{-1}$. We have an augmentation $\epsilon : \mathbb{C}[G] \to \mathbb{C}$, $\sum_g \lambda_g g \mapsto \sum_g \lambda_g$. Its kernel is an ideal.

6. The algebra $D := \mathbb{C}[\varepsilon]/(\varepsilon^2)$ becomes a $*$-algebra by setting $\varepsilon^* = \varepsilon$. It contains a non-zero selfadjoint element whose square is zero.

7. A complex vector space with an antilinear involution becomes a $*$-algebra by giving it zero multiplication.

□

Construction 2.5 (Free $*$-algebras)**.** We have an adjunction

$$F : \mathbf{Set} \leftrightarrows *\mathbf{Alg}^{\mathrm{nu}} : U \ , \tag{2.1}$$

where U takes the underlying set. The left-adjoint sends a set S to the free $*$-algebra $F(S)$ generated by this set. Explicitly, $F(S)$ is the complex vector space with basis the non-empty words in the formal symbols s, s^*, $s \in S$. The two symbols associated to each s are distinct. Multiplication is concatenation. Involution reverses words, exchanges s with s^*, and conjugates coefficients. One easily checks that there is a natural bijection

$$\mathtt{Hom}_{*\mathbf{Alg}^{\mathrm{nu}}}(F(S), A) \xrightarrow{\cong} \mathtt{Hom}_{\mathbf{Set}}(S, U(A)) \ ,$$

given by restriction along $S \to F(S)$. For the inverse, given a function $f : S \to U(A)$, send s to $f(s)$ and s^* to $f(s)^*$, extend to words by multiplication, and then extend complex-linearly. This is the unique $*$-homomorphism extending f. Note that we omit the empty word because a unit is not part of the structure being freely generated. In particular, $F(\varnothing) = 0$. □

Proposition 2.6. *The category* $*\mathbf{Alg}^{\mathrm{nu}}$ *admits all small limits and colimits.*

Proof. In view of the adjunction (2.1) we know that limits in $*\mathbf{Alg}^{\mathrm{nu}}$ are created by $U : *\mathbf{Alg}^{\mathrm{nu}} \to \mathbf{Set}$. For a diagram $A : I \to *\mathbf{Alg}^{\mathrm{nu}}$, its limit is thus represented by

$$L^{\mathrm{alg}} := \left\{ (a_i) \in \prod_{i \in \mathrm{Ob}(I)} U(A_i) \;\middle|\; A(u)(a_i) = a_j \text{ for every } u : i \to j \right\}. \tag{2.2}$$

The operations are coordinatewise. We have a family of projections $(p_i : L^{\mathrm{alg}} \to A_i)_{i \in I}$ given by $p((a_i)) := a_i$ which witnesses L^{alg} as the limit of the family in $*\mathbf{Alg}^{\mathrm{nu}}$.

The empty product is 0.

For colimits, first construct the coproduct of a family (A_i). Form the free $*$-algebra on the disjoint union of their underlying sets. Write $[i, a]$ for the generator corresponding to $a \in A_i$, and quotient by the ideal generated by

$$\begin{gathered} [i, a + b] - [i, a] - [i, b], \qquad [i, \lambda a] - \lambda [i, a], \\ [i, ab] - [i, a][i, b], \qquad [i, a^*] - [i, a]^*. \end{gathered}$$

The resulting algebra, denoted $*_i^{\mathrm{alg}} A_i$, comes with homomorphisms $j_i : A_i \to *_i^{\mathrm{alg}} A_i$. A homomorphism from it to B is exactly a family of homomorphisms $A_i \to B$. The relations impose the algebraic operations within each factor, without imposing relations between distinct factors.

For a general diagram $A : I \to *\mathbf{Alg}^{\mathrm{nu}}$, take this coproduct over its objects and impose the additional relations

$$j_j(A(u)(a)) - j_i(a) = 0 \qquad (u : i \to j,\ a \in A_i).$$

Here imposing relations means forming the quotient by the $*$-ideal generated by the relations. Maps out of the resulting quotient are exactly compatible cocones. This proves its colimit universal property. The empty coproduct is 0. □

Example 2.7. In particular, an equalizer is the subalgebra $\{a \in A \mid f(a) = g(a)\}$ for $f, g : A \to B$, whereas a coequalizer is

$$B / \langle f(a) - g(a) \mid a \in A \rangle_{*\text{-ideal}}.$$

A pullback of $A \xrightarrow{f} C \xleftarrow{g} B$ is the subalgebra $\{(a, b) \in A \times B \mid f(a) = g(b)\}$. □

Recall that an object P in a category is *compact* if $\mathtt{Hom}(P, -)$ preserves filtered colimits. A category is *compactly presented* if it is cocomplete and has a set of compact objects from which every object can be obtained as a filtered colimit.

Proposition 2.8. *The category $*\mathbf{Alg}^{\mathrm{nu}}$ is compactly presented. More explicitly, filtered colimits are created by $U : *\mathbf{Alg}^{\mathrm{nu}} \to \mathbf{Set}$, and every $*$-algebra is a filtered colimit of algebras of the form*

$$F(S)/\langle R\rangle_{*\text{-}ideal}, \qquad S \text{ finite}, \quad R \subseteq F(S) \text{ finite}.$$

Each algebra of this form is compact.

Proof. Let $(A_i)_{i\in I}$ be a small filtered diagram. Its colimit of underlying sets carries a unique $*$-algebra structure making all structure maps homomorphisms: to add or multiply finitely many elements, choose representatives at a common stage and perform the operation there. Scalar multiplication and involution are defined on representatives in the same way. Filteredness makes these definitions independent of the choices. All algebraic identities involve only finitely many elements, so they hold at a common stage and hence in the colimit. This algebra satisfies the colimit universal property, proving the assertion about U.

Now put $P = F(S)/\langle R\rangle_{*\text{-ideal}}$ with S and R finite. A map $P \to \operatorname{colim}_i A_i$ is determined by the images of its finitely many generators. They can be lifted to a common stage A_i. Each of the finitely many relations vanishes at some later stage, so after passing to a single further stage all relations hold. The lifted generators therefore define a map from P. Similarly, two such maps agreeing in the colimit agree at a common later stage, since it suffices to check equality on S. Thus

$$\operatorname*{colim}_i \mathtt{Hom}_{*\mathbf{Alg}^{\mathrm{nu}}}(P, A_i) \xrightarrow{\cong} \mathtt{Hom}_{*\mathbf{Alg}^{\mathrm{nu}}}(P, \operatorname*{colim}_i A_i),$$

and P is compact.

Finally, fix A. Consider pairs (S, R), where S is a finite subset of $U(A)$ and R is a finite subset of the kernel of the evaluation map $F(S) \to A$. Order these pairs by enlarging S and R, using the natural inclusions between free algebras. This is a filtered poset: unions give a common upper bound for any two pairs. The algebras $F(S)/\langle R\rangle_{*\text{-ideal}}$ form a diagram with a compatible cocone to A. Every element of A occurs as a generator at some stage. Every relation among finitely many such generators occurs at a stage after it is added to R. Hence the induced map from the colimit to A is both surjective and injective, and is an isomorphism.

Up to isomorphism, the finitely presented algebras above form a set: take $S = \{1, \dots, n\}$ for $n \geq 0$ and finite subsets of each $F(S)$ as relation sets. Together with Proposition 2.6, this proves compact presentability of $*\mathbf{Alg}^{\mathrm{nu}}$. □

Remark 2.9. Presentability implies an adjoint functor theorem: Let $\mathcal{D}$ be a locally small ordinary category. A functor $F : *\mathbf{Alg}^{\mathrm{nu}} \to \mathcal{D}$ has a right adjoint if and only if it preserves all small colimits.

Example 2.10. The functor $U : *\mathbf{Alg}^{\mathrm{nu}} \to \mathbf{Alg}^{\mathrm{nu}}$ forgetting involution preserves colimits, since algebraic colimits inherit a compatible involution. Here $\mathbf{Alg}^{\mathrm{nu}}$ denotes possibly non-unital complex algebras. The preceding Remark 2.9 gives a right adjoint, explicitly

$$R(B) = B \times \overline{B}^{\mathrm{op}}, \qquad (b, \bar{c})^* = (c, \bar{b}),$$

where $\overline{B}$ is the conjugate complex algebra. The adjunction sends $f : U(A) \to B$ to the $*$-homomorphism $a \mapsto (f(a), \overline{f(a^*)})$. □

2.2 The maximal seminorm

Let A be in $*\mathbf{Alg}^{\mathrm{nu}}$. A *representation* of A is a morphism of complex $*$-algebras $\rho : A \to \mathbb{B}(H)$ for a complex Hilbert space H.

Definition 2.11. *For $a \in A$, define the maximal seminorm of a by*

$$\|a\|_{\max,A} := \sup_{\rho : A \to \mathbb{B}(H)} \|\rho(a)\| \in [0, \infty] \ ,$$

where the supremum ranges over all representations.

We omit A from the subscript when there is no ambiguity. The zero representation ensures that the supremum is taken over a nonempty family; its value may be infinite.

Lemma 2.12. *For all a, b in A and $\lambda \in \mathbb{C}$:*

$$\|a+b\|_{\max} \le \|a\|_{\max} + \|b\|_{\max}, \qquad \|\lambda a\|_{\max} = |\lambda| \, \|a\|_{\max},$$
$$\|ab\|_{\max} \le \|a\|_{\max}\|b\|_{\max}, \qquad \|a^*\|_{\max} = \|a\|_{\max}, \qquad \|a^*a\|_{\max} = \|a\|_{\max}^2.$$

Here products in $[0, \infty]$ are interpreted with the convention $0 \cdot \infty = \infty \cdot 0 = 0$. Moreover, any $*$-homomorphism $f : A \to B$ satisfies

$$\|f(a)\|_{\max,B} \le \|a\|_{\max,A} \tag{2.3}$$

for every a.

Proof. We use that analogous relations hold for $\rho(a)$ and $\rho(b)$ in each representation. For the triangle inequality, every representation $\rho : A \to \mathbb{B}(H)$ satisfies

$$\|\rho(a+b)\| \le \|\rho(a)\| + \|\rho(b)\| \le \|a\|_{\max} + \|b\|_{\max}.$$

Taking the supremum over ρ on the left proves the claim, also when the right-hand side is infinite. No single representation needs to approximate both maximal norms. Similarly,

$\|\rho(ab)\| \le \|\rho(a)\|\,\|\rho(b)\|$ gives submultiplicativity. If either maximal seminorm is zero, its image under every representation is zero, so the product also has maximal seminorm zero, even if the other seminorm is infinite. Homogeneity for $\lambda = 0$ is immediate; the remaining identities follow by taking suprema of the corresponding operator identities. For the second statement, each representation ρ of B gives the representation ρf of A, so the supremum on the left of (2.3) is taken over a subcollection of the representations available on the right. □

The relation

$$\|a^*a\| = \|a\|^2$$

is called the C^*-identity

Example 2.13. For $\mathbb{C}[t]$ with $t = t^*$, evaluation at each $r \in \mathbb{R}$ is a representation. Every nonconstant polynomial is unbounded on $\mathbb{R}$, so

$$\|p\|_{\max} = \infty \quad (\deg p > 0), \qquad \|\lambda 1\|_{\max} = |\lambda|.$$

□

Example 2.14. In $D = \mathbb{C}[\varepsilon]/(\varepsilon^2)$ with $\varepsilon^* = \varepsilon$, every representation kills ε, since $\|\rho(\varepsilon)\|^2 = \|\rho(\varepsilon^2)\| = 0$. The image of 1 is a projection. Evaluation at $\varepsilon = 0$ then shows

$$\|\lambda 1 + \mu\varepsilon\|_{\max} = |\lambda|.$$

Thus finite maximal seminorm need not be a norm.

Likewise, the maximal seminorm of a zero-multiplication $*$-algebra vanishes identically, because $a^*a = 0$. □

Example 2.15. Consider the commutative $*$-algebra

$$B := \mathbb{C}[x, y]/\big(x(1 + y^2) - 1\big), \qquad x^* = x, \quad y^* = y.$$

Evaluations $y \mapsto t$, $x \mapsto (1 + t^2)^{-1}$ for $t \in \mathbb{R}$ show that the canonical map $\mathbb{C}[x] \to B$ is injective: a polynomial in its kernel vanishes at every point of $(0, 1]$. Nevertheless,

$$\|x\|_{\max,\mathbb{C}[x]} = \infty, \qquad \|x\|_{\max,B} = 1.$$

The first equality follows from Example 2.13. For the second, let $\rho : B \to \mathbb{B}(H)$ be a representation and restrict to $\rho(1)H$, on whose orthogonal complement it vanishes. There $\rho(y)$ is selfadjoint and $\rho(x) = (I + \rho(y)^2)^{-1}$ is a contraction, since $\langle v, (I + \rho(y)^2)v\rangle = \|v\|^2 + \|\rho(y)v\|^2$. Evaluation at $x = 1$, $y = 0$ gives the reverse bound. Thus an injective $*$-homomorphism need not preserve the maximal seminorm. □